

# Scientific Report on Banking Campaign Outcome Prediction

## Introduction

The objective of this project is to predict the outcome of direct marketing campaigns of a banking institution, based on phone calls. Specifically, we aim to predict whether a client will subscribe to a term deposit (binary outcome: 'yes' or 'no'). This task is crucial for optimizing marketing strategies, improving client targeting, and ultimately enhancing the success rate of marketing campaigns.

## Data Processing

### Data Exploration

The initial dataset comprises 41,188 instances and 21 features, including client information, campaign details, and social-economic context attributes. Our exploratory data analysis (EDA) revealed no missing values, but certain categorical features contained 'unknown' categories requiring attention. The dataset also exhibits a significant class imbalance, with only **11.27%** of clients subscribing to a term deposit.

### Preprocessing Steps:

- **Handling 'Unknown' Values:** Categorical features with 'unknown' values were addressed by treating 'unknown' as a separate category, ensuring no data loss.
- **Encoding Categorical Variables:** We applied one-hot encoding to transform categorical variables into a machine-readable format, excluding the target variable 'y'.
- **Class Imbalance:** We used SMOTE (Synthetic Minority Over-sampling Technique) and adjusted class weights to tackle the class imbalance issue.
- **Feature Scaling:** Numerical variables were standardized to ensure consistency in scale.
- **Feature Selection:** The 'duration' feature was excluded from the analysis to prevent data leakage, as it is not known before a call is made.

### Feature Selection Using Random Forest

Feature selection is a crucial step in the machine learning pipeline, particularly in tasks involving high-dimensional data. It involves identifying the most informative features that contribute significantly to the predictive power of a model. Random Forest, a popular ensemble learning method, is widely used for feature selection due to its inherent ability to rank features based on their importance.

## Random Forest for Feature Importance

Random Forest consists of a large number of individual decision trees that operate as an ensemble. Each tree in a Random Forest splits nodes based on a subset of features chosen at random, making the model robust against overfitting and providing an unbiased estimate of feature importance. After training, Random Forest can evaluate the importance of each feature in making accurate predictions, offering insights into which features are most influential.

## Methodology

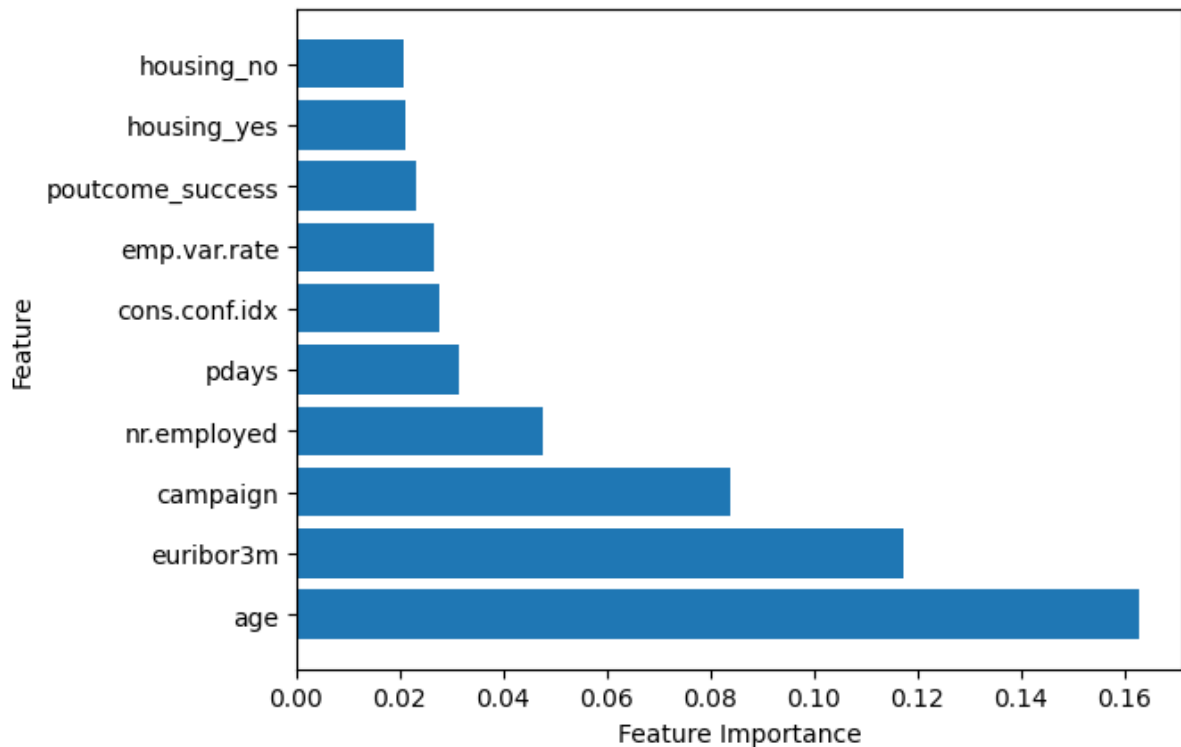
In our project, we utilized Random Forest to identify and rank the features based on their importance for predicting whether a client would subscribe to a term deposit. This approach involved the following steps:

1. **Training a Random Forest Classifier:** We trained a Random Forest classifier on the preprocessed dataset, excluding the 'duration' feature to avoid data leakage.
2. **Extracting Feature Importance Scores:** After training, the model provides a feature importance score for each feature, indicating how valuable each feature was in the ensemble of trees for making predictions.
3. **Ranking Features:** We ranked the features based on their importance scores, allowing us to focus on the most significant features.

## Results

The feature importance analysis revealed the top 10 features as follows:

1. **Age:** Age of the client showed the highest importance, suggesting that clients of certain ages might be more inclined to subscribe to term deposits.
2. **Euribor3m:** The 3-month Euribor interest rate was also highly important, indicating the influence of economic conditions on decision-making.
3. **Campaign:** The number of contacts performed during the campaign for a client was a significant predictor, highlighting the effect of marketing efforts.
4. **Nr.employed:** The number of employees - a proxy for the company's size or economic activity level - was among the top features.
5. **Pdays:** Days since the client was last contacted from a previous campaign also ranked high, suggesting the impact of previous interactions on current decisions.
6. **Cons.conf.idx:** The consumer confidence index, reflecting economic optimism or pessimism, was a key feature.
7. **Emp.var.rate:** The employment variation rate, indicating economic stability, was important.
8. **Poutcome\_success:** The success of the previous marketing campaign was a predictive feature, implying that past success can influence future outcomes.
9. **Housing\_yes:** Having a housing loan appeared to be a significant factor.
10. **Housing\_no:** Conversely, not having a housing loan also provided important predictive power.



## Modelling

In addressing the challenge of predicting the outcomes of banking marketing campaigns, we adopted a rigorous approach involving Logistic Regression and Random Forest models. Our dataset exhibited a notable class imbalance, prompting us to experiment with different configurations such as using class weights and SMOTE (Synthetic Minority Oversampling Technique) to mitigate this issue and enhance model performance. Here, we delve into the detailed analysis of these configurations and their resultant model performances.

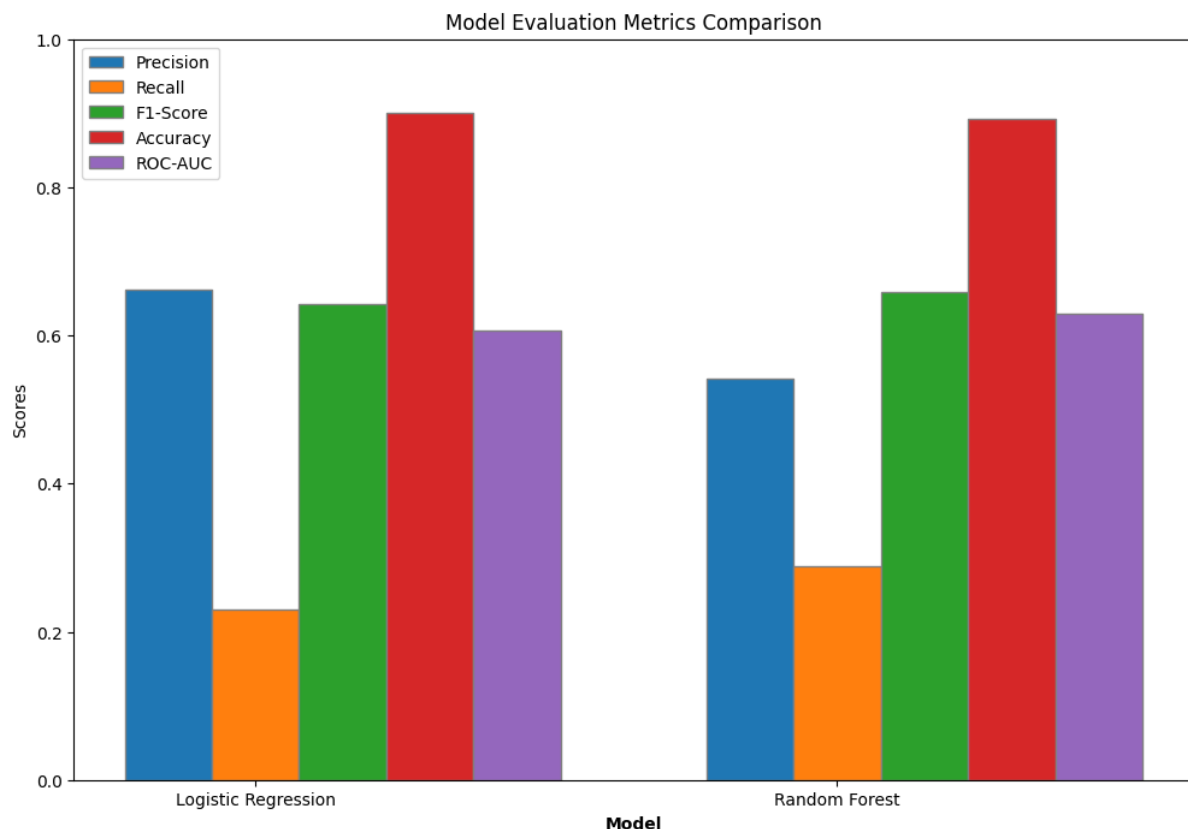
### Configuration A: Baseline Models

#### Logistic Regression

- **Precision:** The model achieved a precision of 66.23%, indicating a relatively high likelihood of accurately predicting positive (subscription) cases. However, this comes with the caveat of a low recall.
- **Recall:** At 22.93%, the model struggled significantly to identify the majority of actual positive cases, reflecting the challenge posed by class imbalance.
- **F1-Score:** The F1-score of 64.30% suggests a moderate balance between precision and recall, albeit skewed towards precision.
- **Accuracy:** High accuracy of 90.01% reflects the model's ability to correctly predict outcomes, heavily influenced by the majority class due to class imbalance.
- **ROC-AUC:** A score of 60.73% indicates limited ability in distinguishing between the classes effectively.

## Random Forest

- **Precision:** Lower precision (54.17%) compared to Logistic Regression, indicating more false positives.
- **Recall:** Improved recall (28.90%) compared to Logistic Regression, but still inadequate in capturing the majority of positive cases.
- **F1-Score and Accuracy:** Slightly higher F1-score (65.89%) and comparable accuracy (89.24%) suggest that Random Forest is somewhat more balanced but still affected by class imbalance.
- **ROC-AUC:** A slightly higher ROC-AUC score (62.90%) than Logistic Regression, indicating a marginally better discriminatory ability.



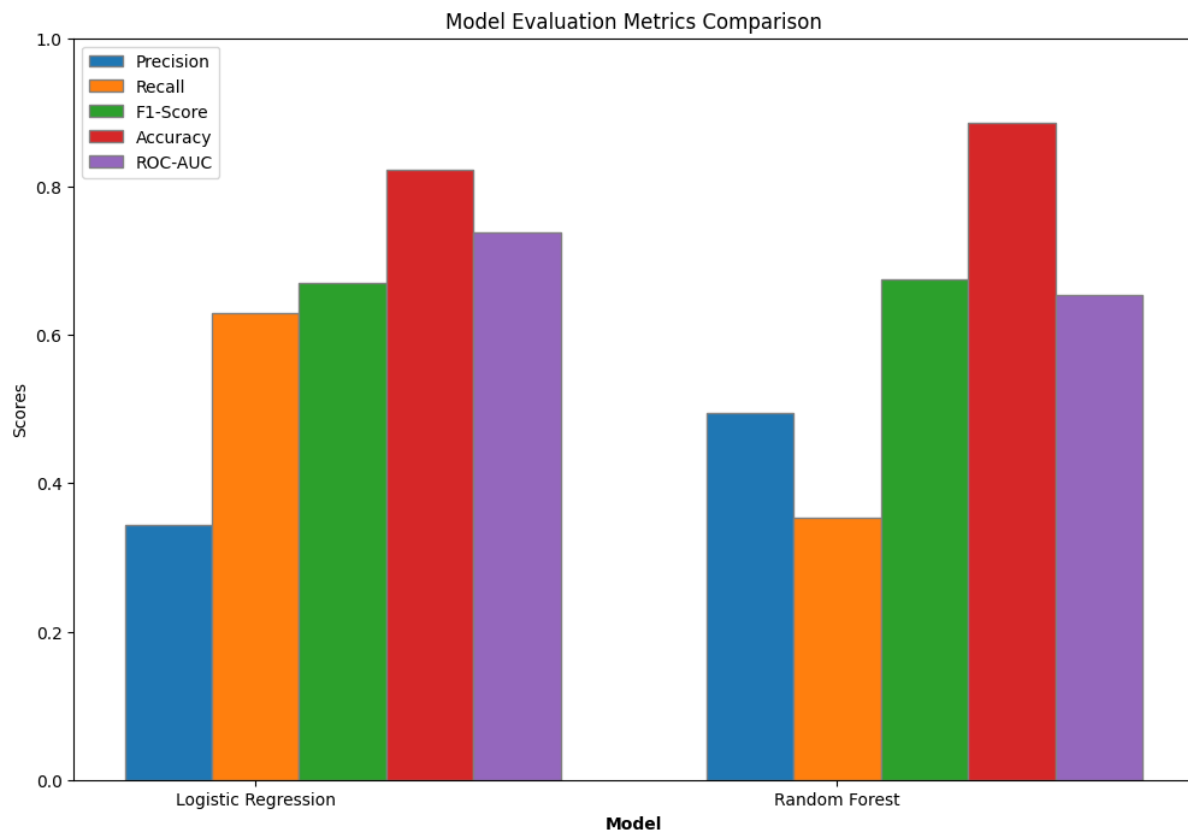
## Configuration B: Adjusting for Class Imbalance

### Logistic Regression with Class Weights and SMOTE

- **Precision & Recall:** A significant shift, with precision dropping to 34.32% but recall increasing to 62.95%. This indicates a strategic trade-off, favoring the model's ability to detect more true positives at the expense of accepting more false positives.
- **F1-Score & ROC-AUC:** Improvements in both metrics (F1-score: 66.92%, ROC-AUC: 73.82%) underscore the enhanced balance and discriminatory power achieved by addressing class imbalance.

## Random Forest with Class Weights and SMOTE

- **Precision & Recall:** Random Forest also exhibits a trade-off, with precision at 49.40% and recall at 35.39%. The model retains a better precision compared to Logistic Regression under the same configuration.
- **F1-Score & ROC-AUC:** The model achieves an F1-score of 67.47% and a ROC-AUC of 65.39%, highlighting its effective balance and improved ability to distinguish between classes after adjustments.



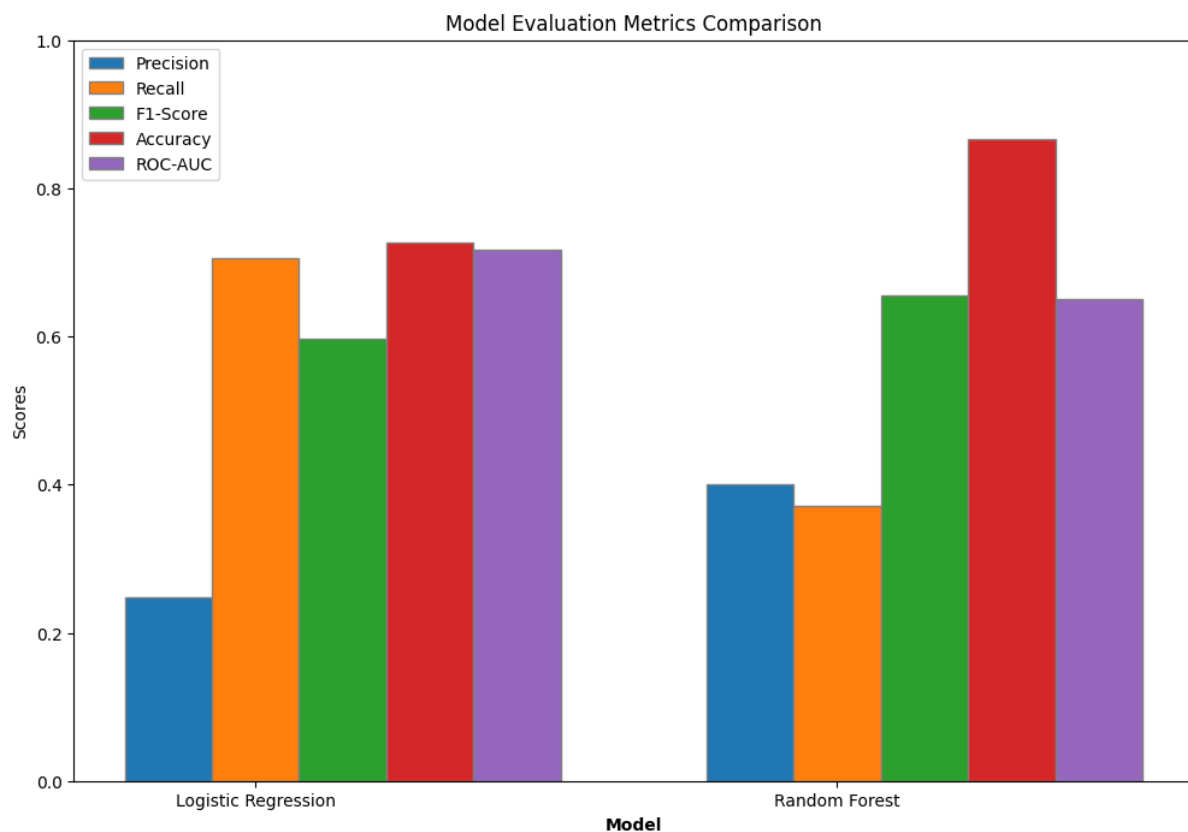
## Configuration C: Important Features with Class Weights and SMOTE

### Logistic Regression

- **Precision & Recall:** A further trade-off is observed, with precision dropping to 24.83% but recall soaring to 70.52%, indicating a focused strategy on maximizing true positive identification.
- **F1-Score & ROC-AUC:** The model's F1-score (59.63%) and ROC-AUC (71.70%) reflect its improved capability in handling imbalanced data, prioritizing recall.

### Random Forest

- **Precision & Recall:** Exhibits a balanced approach with precision at 39.97% and recall at 37.16%, demonstrating an equitable trade-off.
- **F1-Score & ROC-AUC:** The model achieves a robust F1-score (65.50%) and ROC-AUC (65.04%), indicating its consistency in performance across metrics.



## Conclusions from Model Evaluations

- **Impact of Class Imbalance:** Initial models without adjustments for class imbalance leaned towards predicting the majority class, as evidenced by high accuracy but low recall rates.
- **Adjustments for Class Imbalance:** Introducing class weights and SMOTE significantly improved recall for both models, especially for Logistic Regression, albeit at the cost of reduced precision.
- **Feature Selection:** Focusing on important features further enhanced recall, particularly for Logistic Regression, underscoring the value of targeted feature selection in improving model sensitivity to the minority class.
- **Model Comparison:** While both models demonstrated strengths under different configurations, Logistic Regression with class weights and SMOTE, focusing on important features, emerged as particularly effective in identifying positive cases, despite lower precision. Random Forest maintained a more balanced performance across metrics, making it a robust choice for scenarios requiring a balance between precision and recall.

The experimentation underscored the critical importance of addressing class imbalance and carefully selecting features to improve model performance. Through strategic adjustments and targeted feature selection, we were able to significantly enhance model sensitivity to the minority class, thereby improving the overall predictive capability of our models for the banking marketing campaign outcome prediction.

**Note:** Training and evaluation were done using Stratified K-Fold cross-validation with 5 folds, ensuring that each fold was representative of the overall class distribution.

## Conclusion

### Scientific Bottlenecks and Overcoming Them

The primary challenges in this project were the class imbalance and determining the optimal set of features for prediction. We addressed these through SMOTE, class weight adjustments, and empirical feature selection, which significantly improved model performance, particularly for the minority class.

### Algorithm Comparison

- **Logistic Regression** proved valuable for its interpretability and showed considerable improvement in recall for the minority class after applying SMOTE and class weight adjustments.
- **Random Forest** offered robust performance across metrics and benefited from feature selection, highlighting its capability to handle complex relationships in the data.

### Final Thoughts

In conclusion, this project underscores the importance of thorough data preprocessing, the judicious choice of modeling techniques, and the necessity of addressing class imbalance in predictive modeling. Future work could explore more advanced resampling techniques, feature engineering, and ensemble methods to further enhance model performance.